

Ion channel gating

The written notes for the ion channel gating walkthrough.

To print: these notes as a two-column PDF. Already printed the Week 4 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

Leak channels and gated channels

- An ion channel is a membrane protein with a water-filled pore that lets specific ions cross the membrane.
- A leak channel has no gate, so its pore is open all the time. Each leak channel is selective: it lets only certain ions through, and those ions move through it continuously, down their concentration and electrical gradients.
- In a resting neuron most leak channels are potassium leak channels. Potassium leaving through them is the main reason the resting membrane potential is negative.
- Most ion channels are gated. Part of the channel protein, the gate, blocks the pore when the channel is closed. When a specific stimulus acts on the channel, it causes a conformational change, a change in the shape of the protein, that moves the gate out of the pore so ions can pass.
- Leak channels set the resting membrane potential. Gated channels change the membrane potential at a specific time, in response to a specific stimulus, and that change is the signal.

Leak channel

A selective channel with no gate, open all the time.

Gated channel

A channel with a gate that opens only in response to a specific stimulus.

Conformational change

A change in the shape of a protein. In a gated channel it moves the gate out of the pore.

What opens a gate

Channel	Opens when	Where you meet it
Voltage-gated	The membrane potential changes. A voltage sensor, a region of the protein with charged amino acids that sits in the electric field across the membrane, moves when the membrane depolarizes, and that causes a conformational change that opens the gate.	The sodium and potassium channels of axons; calcium channels in axon terminals
Ligand-gated (chemically gated)	A specific ligand binds to a binding site on the channel. The binding site fits only molecules with a particular shape and charge, which is why each channel responds to specific ligands. Binding causes a conformational change that opens the gate, and the channel stays open while the ligand is bound.	Acetylcholine receptors at the neuromuscular junction, where a motor neuron meets a skeletal muscle fiber
Mechanically gated	The membrane is stretched, bent or pressed. The deformation changes the shape of the channel protein and opens the gate. No voltage change or ligand is needed.	Touch receptors in the skin, hair cells of the inner ear, stretch receptors in the walls of blood vessels

- Positive feedback in voltage-gated sodium channels: channels open when the membrane depolarizes, sodium enters and depolarizes the membrane further, and that opens more channels. This is why an action potential, once it reaches threshold, always goes to its full size.
- Ligand-gated channels are the basis of chemical synaptic transmission. One cell releases a neurotransmitter, it diffuses across the synaptic cleft, and it binds to ligand-gated channels on the next cell.
- A drug that binds to the acetylcholine binding sites without opening the channels prevents acetylcholine from binding. The channels stay closed, the muscle fiber does not depolarize, no action potential starts in the muscle, and the muscle does not contract: weakness or paralysis. Some paralytic drugs used during surgery work this way.
- In myasthenia gravis, antibodies attack these acetylcholine receptors and reduce their number, which causes muscle weakness that worsens with repeated use.
- Mechanically gated channels carry out sensory transduction for touch, hearing and the sensing of blood pressure: the mechanical stimulus opens the channels, ions cross the membrane, and the membrane potential changes.

Sensory transduction

The conversion of a stimulus, such as pressure, sound vibration or stretch, into an electrical signal.

The sodium channel has two gates

- Activation gate: closed at rest; opens quickly when the membrane depolarizes.
- Inactivation gate: on the cytoplasmic side of the channel. Open at rest; closes more slowly in response to the same depolarization, and then blocks the pore.
- Sodium can pass only when both gates are open.

State	Activation gate	Inactivation gate	Na ⁺ passes?	Can it open now?
Resting (closed)	Closed	Open	No	Yes, if the membrane depolarizes
Open	Open	Open	Yes, sodium enters rapidly	It is open
Inactivated	Open (or closed briefly during repolarization)	Closed, blocking the pore	No	No, however strong the stimulus

The sequence each time the channel opens

1. At the resting potential, about -70 mV, the channel is in the resting state.
2. As the membrane depolarizes, activation gates begin to open. At threshold, about -55 mV, enough sodium channels are open that sodium entering the cell outpaces potassium leaving it, and the depolarization becomes self-reinforcing.
3. Sodium enters rapidly, because its concentration gradient and the electrical gradient both act inward, and the membrane potential rises to about $+30$ mV. This is the rising phase of the action potential.
4. About a millisecond after the activation gate opens, the inactivation gate closes and blocks the pore from the cytoplasmic side. Sodium entry stops automatically, because the same depolarization that opened the channel also closes it.
5. The membrane repolarizes, mainly because potassium leaves the cell.
6. Only then does the channel return to the resting state: the activation gate closes, then the inactivation gate reopens, and the channel can open again.

Inactivation

The inactivation gate blocking the pore. A channel in the resting state can open when the membrane depolarizes; an inactivated channel cannot open, no matter how strong the stimulus.

Absolute refractory period

The time after an action potential when most sodium channels are inactivated, so no stimulus, however strong, can produce another action potential in that region. It limits how many action potentials a neuron can produce per second, and it is why an action potential travels in one direction along an axon.

Return to the resting state

Recovery from inactivation, which happens only after the membrane repolarizes.

The voltage-gated potassium channel

- It opens in response to the same depolarization as the sodium channel, but it has a single gate and it opens more slowly.
- It opens at about the time the sodium channels are inactivating, and potassium leaving the cell repolarizes the membrane.
- It is also slow to close, so potassium keeps leaving after the membrane potential has returned to its resting value. The membrane potential becomes more negative than rest until those channels close.
- After-hyperpolarization, also called the undershoot, is that period below the resting potential. It contributes to the relative refractory period, when a stronger than normal stimulus is needed to fire the neuron, because the membrane is farther from threshold than usual.

After-hyperpolarization

The period after repolarization when the membrane potential is more negative than rest, because voltage-gated potassium channels are slow to close.

Clinical applications

Patient	What happens	Why
Local anesthetic (lidocaine) before a dental procedure	Within a few minutes the gums and lip on that side are numb	Lidocaine binds inside the pore of voltage-gated sodium channels and blocks it, so sodium cannot enter even when the activation gate opens. The axon cannot produce action potentials, so the pain signal is not conducted to the brain. Lidocaine binds most readily to open and inactivated channels, so it blocks frequently firing nerves most effectively, and small pain fibers are blocked at lower concentrations than large motor fibers.
Severe hyperkalemia	Excitable tissue stops responding; life-threatening arrhythmias in the heart	High extracellular potassium keeps the membrane depolarized, so sodium channels stay inactivated and do not return to the resting state.
Hyperkalemic periodic paralysis	Episodes of severe muscle weakness that start during rest after exercise or after eating potassium-rich foods; blood potassium is often high during an episode; strength is normal between episodes	An inherited mutation in the skeletal muscle sodium channel impairs inactivation. Channels that do not inactivate completely keep letting sodium in and hold the membrane depolarized, which keeps most sodium channels, including the normal ones, inactivated. The muscle fiber cannot produce action potentials, so it cannot contract. Sustained depolarization stops activity rather than increasing it.